

L-S Z Reduction

Chapter 5

(Lehmann - Symanzik - Zimmermann)

We can get particle state

$$|k\rangle = a^\dagger(\vec{k})|0\rangle$$

$$a^\dagger(\vec{k}) = -i \int d^3x e^{i\vec{k}\cdot\vec{x}} \overleftrightarrow{\partial}_0 \varphi(x)$$

$$a(\vec{k}')|0\rangle = 0$$

We require the normalization, such that

$$\langle 0|0\rangle = 1$$

Then the one-particle state will have the

Lorentz invariant normalization

$$\langle k|k'\rangle = \langle 0|a^\dagger(\vec{k}) a(\vec{k}')|0\rangle$$

$$= (2\pi)^3 2\omega \delta^3(\vec{k} - \vec{k}') \langle 0|0\rangle$$

$$= (2\pi)^3 2\omega \delta^3(\vec{k} - \vec{k}')$$

Now we can define a time-independent operator that create a particle localised in $\vec{k}_1$ and near the origin $\vec{x}=0$.

(Recall "minimal wave packet" in QM free particle)

$$a_{\pm}^{\dagger} = \int d^3k f_1(\vec{k}) a^{\dagger}(\vec{k})$$

where $f_1(\vec{k}) \propto e^{-\frac{(\vec{k}-\vec{k}_1)^2}{4\sigma^2}}$

σ is the spread in momentum space.

If we time evolve this wave packet (in Schrödinger Picture), ~~both~~ the ~~two~~ localized particle in $\vec{x}=0$ will spread out at $t \rightarrow \pm\infty$

If we consider two particles

$a_1^{\dagger} a_2^{\dagger} |0\rangle$, at $t \rightarrow -\infty$, these two particles are both spread out the space, but as long as $\vec{k}_1 \neq \vec{k}_2$, they are ^{wildly} separated.

Now Assume it still works for interacting theory.
 we will be back to this point soonish.

$A^+(\vec{k})$ can no longer be time-independent.
(as they will meet and interact).

A suitable initial state is still $(\vec{k}_1 \neq \vec{k}_2)$

$$|i\rangle = \lim_{t \rightarrow -\infty} a_1^+(t) a_2^+(t) |0\rangle$$

but proper normalizations, we can make $\langle i|i \rangle = 1$

Similarly, final states

$$|f\rangle = \lim_{t \rightarrow +\infty} a_1^{*+}(t) a_2^+(t) |0\rangle$$

$$(\vec{k}_1' \neq \vec{k}_2')$$

1 Also take every final momentum different from every initial one, $|\mathbf{k}_{i'} - \mathbf{k}_j| \gg \sigma$. Otherwise $\langle f|i \rangle$ also contains the no-scattering part of $S = 1 + iT$: terms built from the packet overlaps $O_{i'j} \sim e^{-(\mathbf{k}_{i'} - \mathbf{k}_j)^2 / 8\sigma^2}$, which the step on p. 5 drops. (Free theory: p. 5 gives 0, while $\langle f|i \rangle = O_{1'1}O_{2'2} + O_{1'2}O_{2'1}$.)

Now the scattering amplitude is given by

$$\langle f|i \rangle.$$

and we need to find a more useful expression to evaluate $\langle f|i \rangle$.

Note that

(5)

$$\begin{aligned}
a_1^\dagger(+\infty) - a_1^\dagger(-\infty) &= \int_{-\infty}^{\infty} dt \partial_0 a_1^\dagger(t) \\
&= -i \int d^3k f_1(\vec{k}) \int d^4x \partial_0 (e^{ikx} \partial_0 \varphi(x)) \\
&= -i \int d^3k f_1(\vec{k}) \int d^4x e^{ikx} (\partial_0^2 + \omega^2) \varphi(x) \\
&= -i \int d^3k f_1(\vec{k}) \int d^4x e^{ikx} (\partial_0^2 + \vec{k}^2 + m^2) \varphi(x) \\
&= -i \int d^3k f_1(\vec{k}) \int d^4x (e^{ikx} (\partial_0^2 + \cancel{\vec{k}^2} + m^2) \varphi(x) + (\vec{\nabla}^2 e^{ikx}) \varphi(x)) \\
&= -i \int d^3k f_1(\vec{k}) \int d^4x (e^{ikx} (\partial_0^2 - \vec{\nabla}^2 + m^2) \varphi(x) + \text{surface terms (dropped)}) \\
&= -i \int d^3k f_1(\vec{k}) \int d^4x e^{ikx} (-\partial^2 + m^2) \varphi(x)
\end{aligned}$$

In a free field theory $(-\partial^2 + m^2) \varphi(x) = 0$
 as $\varphi(x)$ satisfies the Klein-Gordon Equation.

In an interacting theory, (say) $\mathcal{L}_I = \frac{1}{6} g \varphi^3$

how the E.o.M. gives

$$(-\partial^2 + m^2) \varphi(x) = \frac{1}{2} g \varphi^2$$

$$\text{then } a_1^\dagger(-\infty) = a_1^\dagger(+\infty) + i \int d^3k f_1(\vec{k}) \int d^4x e^{ikx} (\partial^2 + m^2) \varphi(x) \quad \dots (5.11)$$

h.c. of the above yields.

$$a_1(+\infty) = a_1(-\infty) + i \int d^3k f_1(\vec{k}) \int d^4x e^{-ikx} (\partial^2 + m^2) \varphi(x) \quad \dots (5.12)$$

so back to

$$\langle f|i \rangle = \langle 0| a_{1,1}(+\infty) a_{2,1}(+\infty) a_{2,1}^\dagger(-\infty) a_{1,1}^\dagger(-\infty) |0 \rangle$$

we introduce a "Time order" T .

$$\langle f|i \rangle = \langle 0| T a_{1,1}(+\infty) a_{2,1}(+\infty) a_{2,1}^\dagger(-\infty) a_{1,1}^\dagger(-\infty) |0 \rangle$$

¹ T is a definition: it orders by time label and produces no commutator terms. Ordering a against $a^\dagger$ would be normal ordering.

T means anything to the right of it are ordered (rearranged) in time, regardless of how written. ^①
(Left is later time and right is earlier time).

With (5.11) and (5.12)

$$\langle f|i \rangle = \langle 0| T \left((a_{1,1}(+\infty) + i \int \frac{d^4x_1}{(2\pi)^4} e^{-ik_1 x_1} (-\partial_1^2 + m^2) \phi(x_1)) \right. \\ \left. (a_{2,1}(+\infty) + i \int \frac{d^4x_2}{(2\pi)^4} e^{-ik_2 x_2} (-\partial_2^2 + m^2) \phi(x_2)) \right. \\ \left. (a_{1,1}^\dagger(-\infty) + i \int \frac{d^4x'_1}{(2\pi)^4} e^{ik'_1 x'_1} (\partial_1'^2 + m^2) \phi(x'_1)) \right. \\ \left. (a_{2,1}^\dagger(-\infty) + i \int \frac{d^4x'_2}{(2\pi)^4} e^{ik'_2 x'_2} (\partial_2'^2 + m^2) \phi(x'_2)) \right) |0 \rangle$$

² (any $a, a^\dagger$ in above is ~~not~~ zero when acting on $|0\rangle$, or $\langle 0|$) ^②

² $a^\dagger(+\infty)$ is latest, far left: $\langle 0|a^\dagger(+\infty) = 0$, the out-vacuum. $a(-\infty)$ is earliest, far right: $a(-\infty)|0\rangle = 0$, the in-vacuum (p. 6). Exact up to packet-overlap terms (p. 3).

$$= i^{2+2} \int d^4x_1 d^4x_2 d^4x'_1 d^4x'_2 \\ e^{-ik_1 x_1} e^{-ik_2 x_2} e^{ik'_1 x'_1} e^{ik'_2 x'_2} \\ (-\partial_1^2 + m^2)(-\partial_2^2 + m^2)(-\partial_1'^2 + m^2)(-\partial_2'^2 + m^2)$$

$$\langle 0| T \phi(x_1) \phi(x_2) \phi(x'_1) \phi(x'_2) |0 \rangle$$

$$\langle f | i \rangle_n = i^{n+n'} \int d^4 x_1 e^{i k_1 x_1} (-\partial_1^2 + m^2) \dots \int d^4 x_{n'} e^{-i k_{n'} x_{n'}} (-\partial_{n'}^2 + m^2) \dots$$

$$\times \langle 0 | T \underbrace{\varphi(x_1) \dots \varphi(x_n)}_n \underbrace{\varphi(x_{n'+1}) \dots \varphi(x_{n'+n'})}_{n'} | 0 \rangle$$

This is the LSZ reduction formula.

one of the key equations of QFT.

Important additional considerations (Canonical quantization makes these transparent)

But here we are assuming that $\langle f |$ and $| i \rangle$ are described by free particles while $\langle f | i \rangle \neq 0$ due to interaction.

Is it a valid assumption?

The rest is only briefly mentioned during class

(probably skip this part) ^{following} if time is insufficient during class

$\langle 0 | 0 \rangle = 1$ should still be valid. ¹

~~$\langle 0 | 0 \rangle = 0$~~ ?

$$\varphi(x) = \int d^3 k e^{-i k x} a(k) + a^\dagger(k) e^{i k x} \quad \text{2}$$

$\langle 0 | \varphi(x) | 0 \rangle$ should be zero

2 Free field: $\varphi(x) = \int d^3 k [a(k) e^{i k x} + a^\dagger(k) e^{-i k x}]$: the integral is over k .

1 So $|0\rangle$ must be both the in-vacuum and the out-vacuum: $\langle 0_{\text{out}} | 0_{\text{in}} \rangle = 1$. The vacuum is stable, the unique ground state of a time-independent H , with $P^\mu |0\rangle = 0$. This fails in a time-dependent external field, which creates particles.

$$\varphi(x) = \exp(-iP^\mu x_\mu) \varphi(0) \exp(+iP^\mu x_\mu).$$

$$\langle 0 | \varphi(x) | 0 \rangle = \langle 0 | \varphi(0) | 0 \rangle \quad \text{as } P^\mu | 0 \rangle = 0 \\ = \text{const (I.I.)}$$

we want $a^\dagger(\vec{k}) | 0 \rangle$ to create a single particle state, not a linear combination of 0, single and multiple particle state at $t = \pm \infty$. ①

To satisfy $\langle 0 | \varphi(x) | 0 \rangle = 0$ if we have $\langle 0 | \varphi(x) | 0 \rangle = v$

we redefine $\varphi(x) \rightarrow \varphi(x) - v$

so the new field $\varphi(x)$ has.

$$\langle 0 | \varphi(x) | 0 \rangle = 0. \quad \text{as a scalar, we can shift it.}$$

1 Needed only as $t \rightarrow \pm \infty$, but the limit cannot enforce two of them: the vacuum and one-particle parts of $a_1^\dagger(t) | 0 \rangle$ never decay (p. 10). So $v = 0$ and $\langle p | \varphi(0) | 0 \rangle = 1$ are set by hand. P^μ annihilates bra and ket, so v does not depend on x : $v = 0$ at $\pm \infty$ means $v = 0$ everywhere.

Then $\langle p | \varphi(x) | 0 \rangle = \langle p | e^{-iP^\mu x_\mu} \varphi(0) e^{+iP^\mu x_\mu} | 0 \rangle$
a one-particle state.
 $= e^{-ipx} \underbrace{\langle p | \varphi(0) | 0 \rangle}_{\text{this is a number.}}$

we want $\langle p | \varphi(x) | 0 \rangle = 1$ but ~~they~~ it may not, it is a number depends on I.I. function of P^μ , which is $P^2 = -m^2$ and other parameters. ($x^\mu = 0$ here).

for $\langle p | \varphi(0) | 0 \rangle = Z$, we can again¹

rescale $\varphi(0) \rightarrow \varphi(0)/Z$.
(renormalize).

¹ Here Z is the matrix element itself. Elsewhere (e.g. Peskin & Schroeder) this matrix element is called $\sqrt{Z}$, with Z the residue of the propagator pole. That residue is the Z_φ on p. 10, so $Z_\varphi = Z^2$ in this page's notation.

so we have so far

$$\langle p | \varphi(0) | 0 \rangle = 1, \quad \langle 0 | \varphi(0) | 0 \rangle = 1 \quad (2)$$

² Should read $\langle 0 | \varphi(0) | 0 \rangle = 0$: the shift $\varphi \rightarrow \varphi - v$ on p. 7 enforces it.

What about $\langle p, n | \varphi(x) | 0 \rangle$?

n -particle state with total momentum p

we want it to be zero.

$$\begin{aligned} \langle p, n | \varphi(x) | 0 \rangle &= e^{-ipx} \langle p, n | \varphi(0) | 0 \rangle \\ &= e^{-ipx} A_n(p). \end{aligned}$$

$A_n(p)$ is a function of 2.1. of various particle momentum product.

Actually, we really only need

$\langle p, n | a^\dagger(\pm\infty) | 0 \rangle = 0$; such that our assumption on the initial/final states are reasonable.

Let's be slightly more serious.

⑨

We define the $|p, n\rangle$ state more properly,

$$|\psi\rangle = \sum_n \underbrace{\int d^3p \psi_n(\vec{p})}_{\text{total wave packet}} |p, n\rangle.$$

We want $(+ -) \pm$ to be taken later

$$\langle \psi | a_1^\dagger(t) | 0 \rangle$$

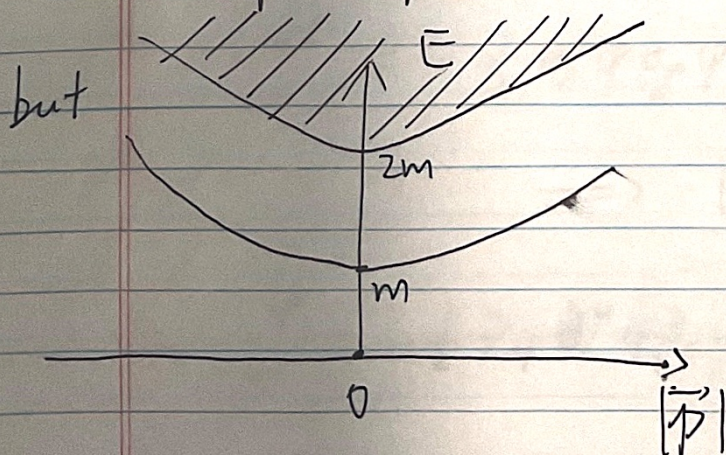
$$= -i \sum_n \int d^3p \psi_n^*(\vec{p}) \int d^3k f_1(\vec{k}) \int d^3x e^{ikx} \frac{\partial}{\partial t} \langle p, n | \phi(x) | 0 \rangle$$

$$= -i \sum_n \int d^3p \psi_n^*(\vec{p}) \int d^3k f_1(\vec{k}) \int d^3x e^{ikx} e^{-ipx} A_n(p)$$

$$= \sum_n \int d^3p \psi_n^*(\vec{p}) \int d^3k f_1(\vec{k}) \int d^3x (k^0 + p^0) e^{i(k-p)x} A_n(p)$$

$$= \sum_n \int d^3p \psi_n^*(\vec{p}) f_1(\vec{p}) (2\pi)^3 (k^0 + p^0) e^{-i(k^0 - p^0)t} A_n(p) \quad (5.23)$$

here $p^0 = (\vec{p}^2 + M^2)^{1/2}$, $k^0 = (\vec{p}^2 + m^2)^{1/2}$



we know then

$$M \geq 2m > m$$

thus, 5.23 oscillates fast at $t \rightarrow \pm\infty$.

1 Sort by the mass M of the intermediate state. Vacuum: phase e^{-imt} , never decays, so $v = 0$. $M = m$: phase $\equiv 1$, the one-particle packet at every t . $M \geq 2m$: oscillates, $\rightarrow 0$. For $a_1(t)|0\rangle$ the one-particle part vanishes identically, so $a_1(\pm\infty)|0\rangle = 0$ (p. 5).

by Riemann-Lebesgue lemma,
the right-hand side vanishes as $t \rightarrow \pm\infty$

1 Why only $n \geq 2$ decays:
 $\langle s | a_1^\dagger(t) | 0 \rangle \propto e^{iK_s t}$,
 $K = H - \sqrt{\mathbf{P}^2 + m^2}$.
 Averaged over a packet this
 decays only if K varies: not
 for the vacuum ($K = -m$) or
 one particle ($K \equiv 0$); yes for
 $n \geq 2$, whose group velocity
 differs. Same reason the m in
 $a^\dagger$ must be the physical mass.

Physically: one particle wave packet $a^\dagger | 0 \rangle$ spread out differently than

$|p, n\rangle$
~~there~~ they are unlikely to overlap via infinite propagation
 (at least by constructing one-particle wave packet
 carefully).

LSZ formula is good so long as

$$\langle 0 | \varphi(x) | 0 \rangle = 0 \quad \text{and} \quad \langle k | \varphi(x) | 0 \rangle = e^{-ikx}.$$

this means shift & rescale an original Lagrangian to
 a new one. e.g.

$$\mathcal{L} = -\frac{1}{2} \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} m^2 \varphi^2 + \frac{1}{6} g \varphi^3$$

$\Rightarrow$ (Will ~~truth~~ formally introduce
 renormalization later).

$$\mathcal{L} = -\frac{1}{2} Z_\varphi \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} Z_m m^2 \varphi^2 + \frac{1}{6} g Z_g \varphi^3 + Y \varphi.$$

we can identify $Z_m m^2 / Z_\varphi$ with the physical mass ~~need~~ and other
 "physically"